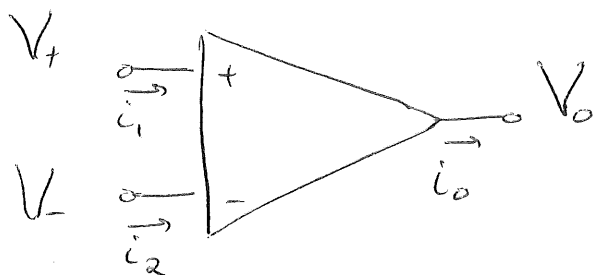


* Ideale Op-Amp



① $i_1 = i_2 = 0 \text{ A}$

② $V_+ = V_-$

* Doelwit: doen wisk. bewerkingen.

① $V_o = -K_I \times V$ ("Inverting")

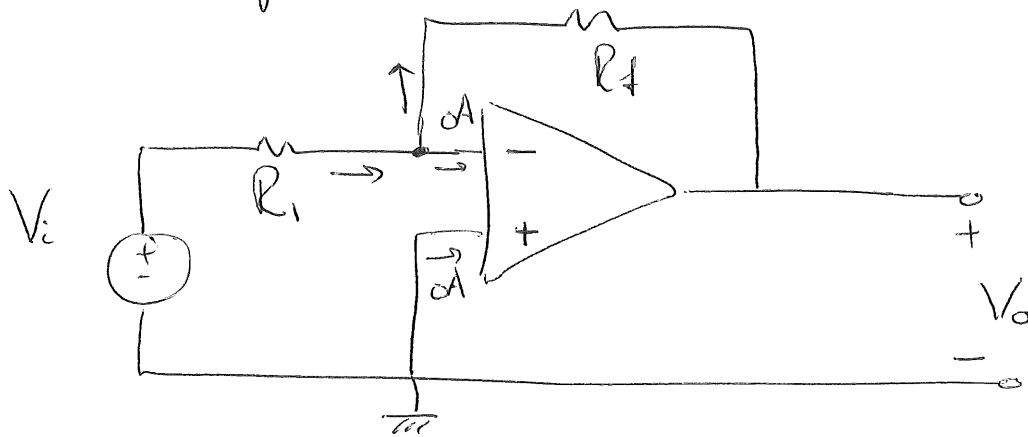
② $V_o = +K_{NI} \times V_i$ ("Non-Inverting")

③ $V_o = +V_i$ (Buffer).



① Omkeer versterker
 ("Inverting amplifier")

② NB: Kompleksies
 van bringbaar!



① $V_- = V_+ = 0V.$ / $i_1 = i_2 = 0A.$

② KCL by V_- :

$$\frac{V_i - V_-}{R_i} = \frac{V_- - V_o}{R_f}$$

$$\therefore \frac{V_i - 0}{R_i} = \frac{0 - V_o}{R_f}$$

$$V_o = -\frac{R_f}{R_i} \times V_i$$

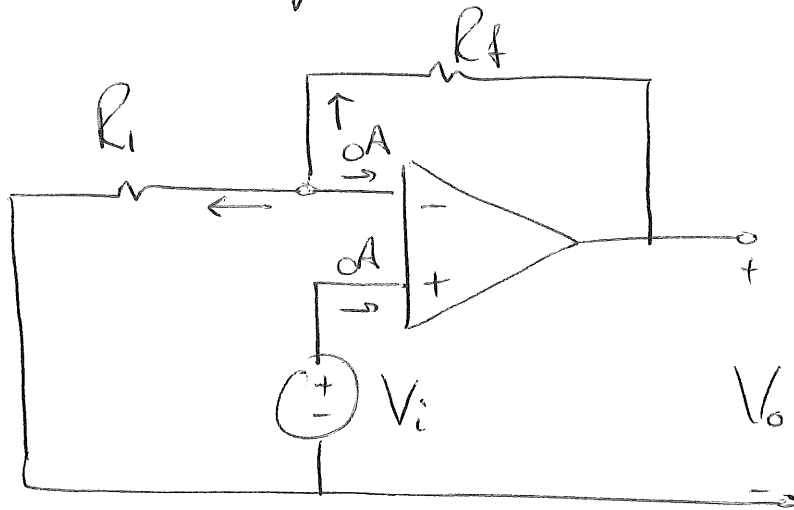
* $\left(-\frac{R_f}{R_i}\right)$ is die omkeer
 versterking van inset V_i .

(2)

Nie-omkeer Versteker

(3)

("Non-inverting amplifier")



* Let op na verskillende in konneksies tussen omkeer en nie-omkeer versterker bringbare.

$$\textcircled{1} \quad V_- = V_+ = V_i \quad / \quad i_1 = i_2 = 0A$$

\textcircled{2} KCL by V_- :

$$\frac{(V_-)}{R_1} + \frac{(V_-) - V_o}{R_f} = 0$$

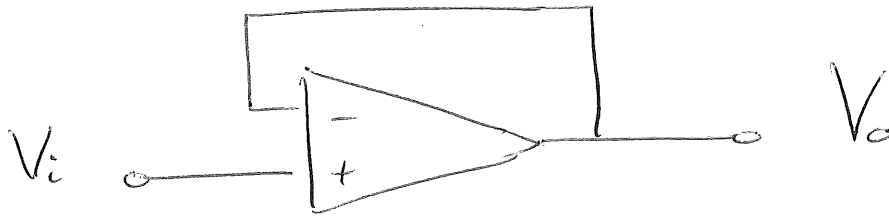
$$\therefore \frac{V_i}{R_1} + \frac{V_i - V_o}{R_f} = 0$$

$$\therefore V_o = \left(\frac{R_f}{R_1} + 1 \right) \times V_i$$

* $\left(\frac{R_f}{R_1} + 1 \right)$ is die nie-omkeer versterking van inset V_i .

③ → Buffer

④



$$V_o = V_i$$

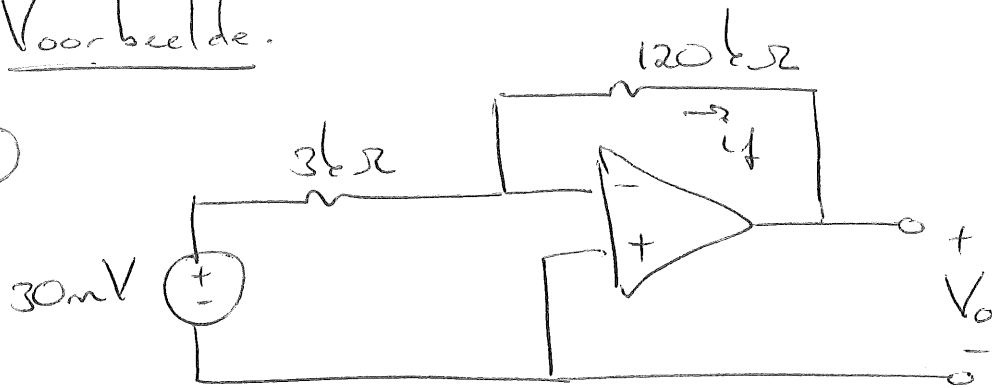
- * Doel is om uitset te isoleer van inset.
- * Dit step is buffer tussen fases in uitgebreide kringbaan.



Voorbeelde.

5

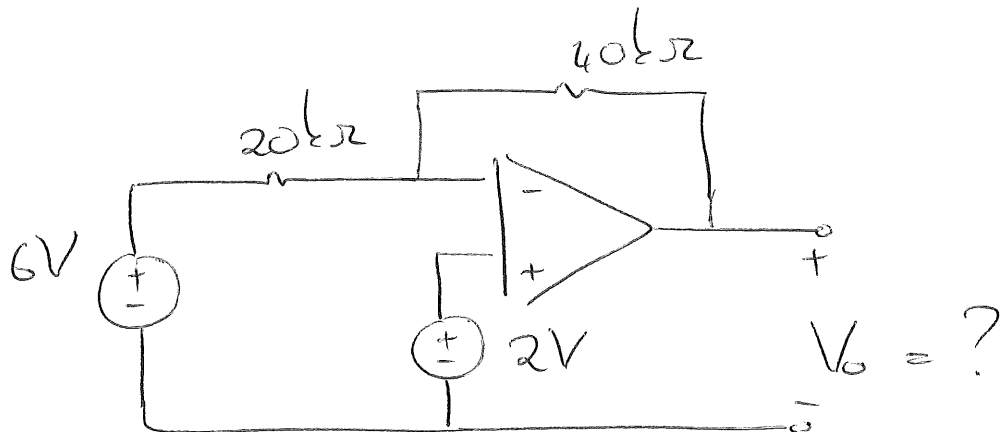
1



$$V_o = ?$$

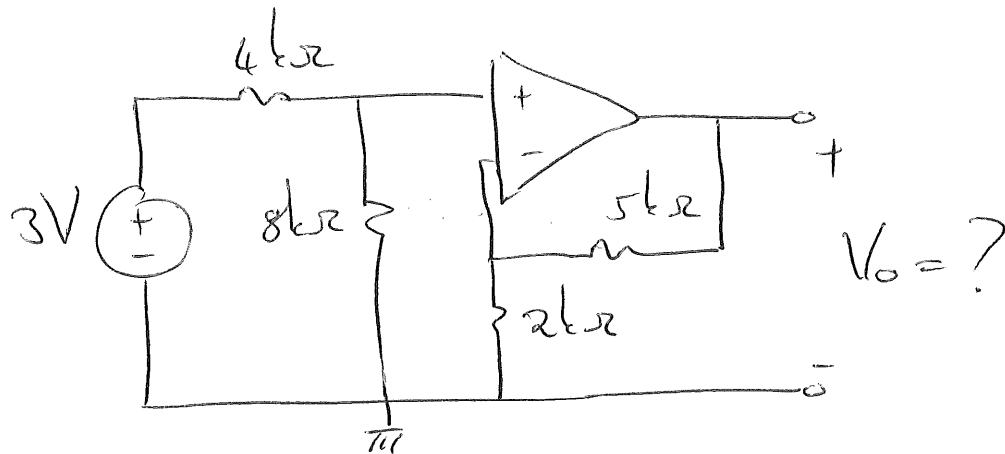
$$i_f = ?$$

2



$$V_o = ?$$

3



$$V_o = ?$$